

RoboMix DAPT — Data Collection Plan

Goal: build a **text & code corpus** for a model that takes a natural-language mission and outputs **(1) task planning** → **(2) task allocation** → **(3) per-robot task planning** → **(4) action-function selection**.

GLOBAL RULES (apply to all 3 categories)

- **Licenses:** collect ONLY MIT / BSD / Apache-2.0 / CC-BY. Anything else (GPL, CC-BY-NC, NoDerivs, no-license) → flag, do NOT collect.
- **Do NOT collect** trajectory / image datasets (OXE, BridgeData, DROID, RoboMIND...) — not text, useless for a text LLM.
- **Contamination:** exclude any text overlapping our evaluation scenario (subway-station fire). Keep the corpus clean.
- **Tokens:** count with the Qwen3 tokenizer. **Target mix:** Cat 1 = 60–70% · Cat 2 = 15–25% · Cat 3 = 10–15% (mandatory).
- **Output — JSONL, 1 record = 1 document:** {id, source, category, tier, license, url, lang, text, n_tokens, collected_at}

Category 1 — General Robot Data (corpus tier A)

serves: (4) action functions + foundation for (1)

What	General robotics knowledge, APIs and code — ROS2 / ROS1 / MoveIt2 / Nav2 official docs; license-cleared robotics GitHub code; planning-as-code (the code side of Code-as-Policies / ProgPrompt / Instruct2Act).
Collect	Async HTML scraping for docs (main content only); GitHub API for repo contents.
Clean	Strip HTML boilerplate (nav/footer/sidebar) → keep code blocks intact → drop auto-generated duplicate pages → remove exact + near-duplicates → scrub API keys/secrets from code.

Category 2 — HMRS Data (corpus tier B)

serves: (1) task planning + (2) allocation + (3) per-robot planning

What	Multi-robot task decomposition, allocation and coordination text — RoCoBench-Text (almost the only clean multi-robot text source), EMOS / Habitat-MAS embodiment-aware allocation descriptions, SMART-LLM, instruction→plan pairs.
Collect	Extract text from benchmark / GitHub repos. For DAPT use only the plan/solution side ; keep full instruction→plan pairs tagged separately (reserved for Stage-2 SFT).
Clean	Normalize episodes/dialogues to plain text → remove simulator logs/noise (keep reasoning + allocation text) → de-duplicate → preserve scenario diversity → run the evaluation-contamination check.

Category 3 — URDF & Robot Specs (corpus tier D, mandatory)

serves: (2) embodiment-aware allocation + feasibility for (3)/(4)

What	Physical capabilities (payload, reach, DOF, joint limits, speed). Priority fleet: Unitree G1 & Go2, AgileX Ranger Mini. URDF / Xacro / SDK + vendor manuals & datasheets.
Collect	URDFs via robot_descriptions.py + official GitHub. Vendor manuals (Unitree / AgileX): do NOT scrape or redistribute — download the PDF, extract text locally, internal-training use only.
Clean	Parse URDF <link mass>, <inertia>, <joint type/limit>, link sizes → convert to natural-language capability sentences (EMOS ‘Robot Resume’ style) → for PDFs strip headers/footers and turn spec tables into readable text → unify units → drop duplicate spec tables → tag the fleet robots.

DEFINITION OF DONE. Zero non-allowlisted licenses · contamination check passed · deliver corpus_clean.jsonl + corpus_stats.md (tier / token / license breakdown). **Workflow:** start with a one-source-per-category end-to-end pilot → review with me → then scale.

Tier letters (A / B / D) follow our master corpus taxonomy in robomix_dapt_corpus.md. Only tiers A, B and D are in scope for this task — other tiers in the master survey are not collected here, so the absence of a ‘tier C’ is expected, not an error.